

OUTLINE ARTIKEL ILMIAH

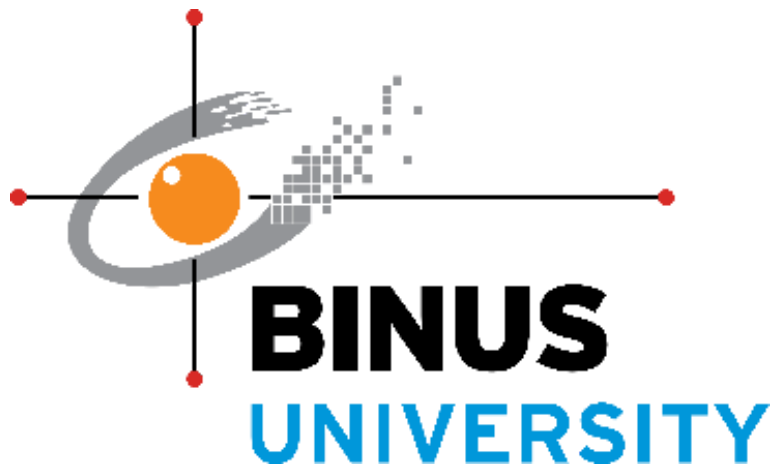
Konversi Teks Tak Terstruktur Menjadi Data Geospasial: Pendekatan Hibrida SLM untuk Menangani Anomali Penulisan Alamat Lokal

From Unstructured Text to Geospatial Data: Handling Localized Address Variances using Hybrid SLM-Rule Based Systems

Topik: Artificial Intelligence

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1. INTRODUCTION

1.1. Background and Problem Identification

Accurate geospatial data is essential to the architecture of many services such as financial, e-commerce, and logistics. However, in Indonesia, address data is known and marked by the characteristic of high variability and structural inconsistency [8]. Not just in Indonesia, non-standardized address systems in developing countries often suffer from spelling errors, ambiguous location descriptions, irregular component ordering, and missing hierarchy information, as noted by Singhal et al. [1]. Some addresses are commonly written with random abbreviations (e.g., “Klp Gdng” for “Kelapa Gading”), local slang, or incomplete details.

Traditional rule-based methods, such as regular expressions, tend to fail in capturing the semantic context needed to parse such noisy, unstructured text. This limitation is documented in a research about cross-country address parsing, notably in Deepparse [2][9], which demonstrates that rule-based and regex-driven systems suffer significantly in the presence of linguistic noise.

The emergence of Large Language Models (LLMs) offers a promising solution due to their strong semantic understanding. However, deploying massive general-purpose LLMs for high-volume data is costly, slow, and raises data privacy risks. LLMs are also prone to "hallucinations", sometimes generating non-existent entities, as shown in recent study AddrLLM [3], where hallucinated or abnormal addresses directly impacted logistics operations.

This creates a research gap for methods that balances the linguistic adaptability with the deterministic accuracy required for data mining. Recent studies indicate that Small Language Models (SLMs) can achieve competitive domain-specific performance with far lower computational cost [5][6], making them strong candidates for low-resource address normalization tasks.

1.2. Regional Context: Challenges in Jakarta

This research concentrates on Jakarta as a heavily populated metropolitan area who possesses a complex administrative structure with numerous neighborhood associations (RT/RW) and sub-districts (Kelurahan), often located within informal settlements. Economically, the region relies heavily on fast last-mile delivery. However, this issue brings out a major bottleneck with ambiguous and noisy address data in Jakarta, driving up to delivery failure rates and operational expenses [11][12]. Similar issues were observed before in AddrLLM [3], where incomplete addresses increased rerouting costs and delays [12]. Thus, converting unstructured addresses into organized and extractable data is an essential economic necessity for strengthening Jakarta’s digital logistics ecosystem.

1.3. Proposed Solution and Expected Contribution

This research proposes a hybrid inference pipeline that integrates a finely-tuned Small Language Model (SLM) with a Rule-Based Knowledge Retrieval system to tackle these challenges. We want to propose a "Proof of Sufficiency" argument: showing that LLMs that use a lot of resources are not needed for addressing parsing tasks. This claim is in line with what recent SLM research has found [5][6], which shows that models with less than 10 billion parameters can do well on specific tasks while keeping inference latency low. By fine-tuning a lightweight SLM (~8B parameters) to handle linguistic variances, unstructured address and pairing it with a deterministic validator, this study offers a scalable and cost-effective framework.

The expected output is an automated pipeline capable of parsing informal texts into standardized geospatial JSON formats. This research contributes to the field by providing a hallucination mitigation strategy specifically tailored for low-resource environments, making it possible to process massive datasets locally with high accuracy and minimal computational overhead.

2. PROBLEM STATEMENT

Based on the background outlined above, the research problems addressed in this study are formulated as follows:

1. **Inefficiency of Traditional Parsing:** How can unstructured and noisy addresses texts in Indonesia (specifically Jakarta) be effectively standardized when conventional rule-based methods fail to capture semantic context? This limitation is well-documented in Deepparse [2] and supported by noise categories from Yin et al. [4].
2. **Resource Constraints vs. Accuracy:** Can a fine-tuned Small Language Model (SLM) with limited parameters (e.g., 8B) achieve parsing accuracy comparable to larger LLMs while maintaining deployment efficiency? Existing SLM literature suggests that compact models can rival LLMs in domain-specific tasks when properly fine-tuned [5][6].
3. **Hallucination Risks:** How can deterministic validation be combined with probabilistic SLM outputs to reduce geospatial hallucinations and preserve data integrity for downstream mining? This is motivated by findings in address-related hallucination research such as AddrLLM [3].

3. SCOPE and LIMITATIONS

To ensure the feasibility and depth of the analysis, this study is bounded by the following scope:

1. **Geospatial Coverage:** The research focuses exclusively on address data within **DKI Jakarta**. Addresses from surrounding cities (e.g., Depok, Tangerang, Bekasi) are considered out-of-scope and automatically filtered by the validation module.
2. **Model Architecture:** The study utilizes **Small Language Models (SLMs)** with parameters under 10 billion (specifically Llama-3-8B-Instruct fine-tuned using the QLoRA technique), optimized for low-resource deployment. Comparisons with closed-source LLMs (e.g., GPT-4) are limited to cost and efficiency, not full performance benchmarking.
3. **Input Data Characteristics:** The system is designed to process **informal Indonesian address text**, including local abbreviations, slangs (*bahasa gaul*), and unstructured formats. It does not cover handwriting recognition (OCR) or voice-to-text inputs.
4. **Output Format:** The standardized output is strictly limited to a predefined JSON schema containing: street name, building number, RT/RW, sub-district (*Kelurahan*), district (*Kecamatan*), city, province, and postal code. Geocoding (converting addresses to latitude/longitude coordinates) is outside the scope of this paper.

4. RESEARCH OBJECTIVE and SIGNIFICANCE

4.1 Research Objective

The primary goal of this study is to **propose and implement** a novel, low-resource framework for address normalization. Specifically, this research aims to:

1. **Propose a Hybrid Methodology:** To introduce a comprehensive approach that combines the linguistic adaptability of Small Language Models (SLMs) with the factual rigidity of Rule-Based validation.
2. **Develop and Implement the Pipeline:** To technically **engineer and deploy** a functional pipeline using a fine-tuned Llama-3-8B model augmented with a Python-based Knowledge Retrieval module.
3. **Empirically Validate Performance:** To **test and demonstrate** the system's capability in mitigating hallucinations and handling localized address variances through rigorous stress-testing on Jakarta datasets.

4.2 Significance of the Study

The findings and methodologies presented in this research offer significant contributions to both the academic community and industrial applications:

1. **Theoretical Contribution (Proof of Sufficiency)**

This study challenges the prevailing "*bigger is better*" paradigm in Natural Language Processing. It provides empirical evidence that **Small Language Models (SLMs)**, when augmented with domain-specific knowledge retrieval, are sufficient for complex information extraction tasks. This contributes a novel framework for **hallucination mitigation** in low-resource settings, offering an alternative to relying solely on probabilistic large models [10].

2. **Practical Implication for Industry**

For Indonesia's rapidly growing logistics and e-commerce sectors, this research offers a **scalable and automated solution** to address the chronic issue of ambiguous address data. By converting noisy text into structured formats, the proposed pipeline can directly reduce last-mile delivery failures and optimize route planning efficiency [12].

3. **Economic and Operational Efficiency**

Unlike solutions dependent on commercial APIs (e.g., GPT-4) which incur high recurring costs and latency, the proposed hybrid pipeline is designed for **edge AI deployment**. It enables startups and SMEs to perform high-volume data cleaning locally on consumer-grade hardware, ensuring data privacy and significantly lowering operational expenses [10].

4. **Alignment with Sustainable Development Goals (SDGs)**

Beyond its technical contributions, this research directly supports several United Nations Sustainable Development Goals (SDGs). By improving address data quality and reducing failed deliveries, the system contributes to *SDG 9: Industry, Innovation, and Infrastructure* and *SDG 11: Sustainable Cities and Communities*. Increased logistical efficiency also supports *SDG 8: Decent Work and Economic Growth*, while structured geospatial data strengthens institutional operations in alignment with *SDG 16: Peace, Justice, and Strong Institutions*.

5. METHODOLOGY

This study adopts a quantitative experimental approach focusing on Data-Centric AI and Hybrid Systems Engineering. The methodology is structured into two main phases: Data Collection & Construction, and System Design & Development.

5.1 Data Collection and Construction Strategy

The primary challenge in address normalization research is the scarcity of large-scale, privacy-compliant public datasets. To address this, we employed a **synthetic data generation** strategy combined with secondary geospatial verification.

5.1.1 Reference Database Acquisition (Secondary Data)

The ground truth for Jakarta's geospatial hierarchy was acquired from an open-source repository [7] ([teguh02/Wilayah-Indonesia-Beserta-Kode-Pos](#)). We filtered and curated this secondary dataset to isolate the **DKI Jakarta** region, resulting in a clean reference corpus containing 267 verified sub-districts (*Kelurahan*) complete with their associated districts (*Kecamatan*), municipalities (*Kota/Kabupaten*), and postal codes. This database serves as the deterministic "Knowledge Base" for our validation module.

5.1.2 Synthetic Training Data Generation Strategy

To address the scarcity of privacy-compliant address datasets, we employed a **"Reverse-Generation" strategy** combining Generative AI with deterministic noise injection. The training corpus was constructed through two parallel streams:

1. **Seed Generation via LLM:** We utilized GPT-4 to generate initial diverse sentence structures, mimicking natural language prompts used by e-commerce users (e.g., *"tolong kirim ke..."*).
2. **Stochastic Noise Injection Module:** To ensure geospatial validity, we developed a Python-based module that systematically degrades clean ground-truth data from the secondary dataset that we have (the ground truth). This module applies three specific degradation rules:
 - **Lexical Substitution:** Mapping standard administrative terms to a dictionary of local colloquialisms and extreme abbreviations (e.g., *"Kelapa Gading" → ["Klp Gdng", "Klpa Gading"]*).
 - **Structural Permutation:** Randomly shuffling the canonical order of address components (e.g., placing postal codes before street names) to force the model to learn semantic context rather than positional heuristics.
 - **Character-Level Perturbation:** Introducing random character insertions or deletions to simulate human typographical errors.

This dual-strategy yielded a dataset of **6.000 pairs**, mapping chaotic raw text inputs to standardized JSON outputs, ensuring the model learns robust features while maintaining zero exposure to real user PII. (*tambahin figure bkin aj*)

5.2 System Design and Development

The proposed solution is engineered as a **Hybrid Inference Pipeline** that synergizes probabilistic neural networks with deterministic rules.

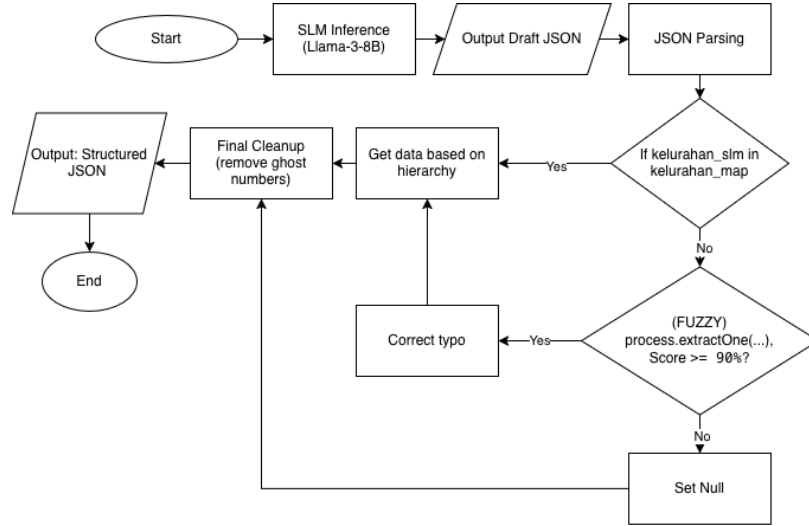


Fig. System Design

5.2.1 Model Architecture (The Probabilistic Component)

We utilized the Llama-3-8B-Instruct as the backbone SLM. The model was optimized using **Parameter-Efficient Fine-Tuning (PEFT)** techniques, specifically **QLoRA (Quantized Low-Rank Adaptation)**. QLoRA is widely recognized as a leading method for low-memory fine-tuning of large and small models [6].

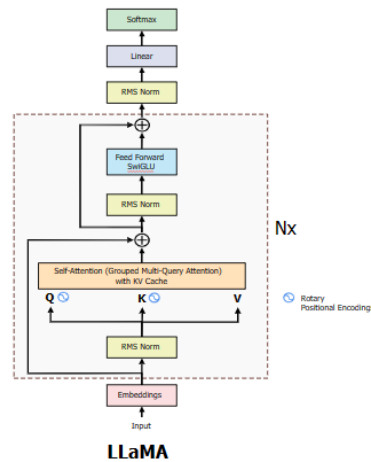


Fig. Llama-3 Architecture that uses Grouped Multi-Query Attention (GQA) dan SwiGLU for efficient computation in low resource hardware.

The model was loaded in 4-bit precision to accommodate the memory constraints of **widely accessible cloud infrastructure** (specifically, a standard **NVIDIA T4 GPU with 16GB VRAM**, available via Google Colab Free Tier). This setup was deliberately chosen to demonstrate the feasibility of deploying the pipeline on zero-cost hardware, leveraging LoRA adapters to extract entities into a JSON draft schema before validation.

5.2.2 Knowledge Retrieval Validator (The Deterministic Component)

To mitigate the hallucination risks inherent in SLMs, we developed a post-processing validation module written in Python. This module implements a **Cascading Verification Protocol**:

1. **Dictionary Lookup:** A predefined map handles known irregular abbreviations (e.g., mapping "*Depok*" to null for scope enforcement).
2. **Database Matching:** If the entity is not found in the dictionary, the system performs an exact lookup against the **Reference Database** (acquired in phase 5.1.1). A successful match automatically retrieves and auto-completes the parent hierarchy (Kecamatan, City, Province).
3. **Fuzzy Logic Correction:** If an exact match fails, the system employs a **Levenshtein Distance** algorithm. It calculates similarity scores against valid database entries. Matches exceeding a strictly defined threshold (Score $\geq \sim 90\%$) are **auto-corrected** (e.g., "*Cikine*" $\rightarrow$ "*Cikini*"). Inputs failing this threshold are deemed invalid, and their hierarchical fields are set to null.
4. **Final Sanitization:** The validated JSON undergoes a final logic check to remove "ghost numbers," specifically stripping the 'nomor' (house number) field if it replicates the postal code, ensuring cleanliness.

5.2.3 Analysis and Evaluation

System performance is evaluated through a comprehensive **Stress-Testing Scenario** using a curated diverse test set. This set specifically targets edge cases, including anomalous inputs (localized slang, extreme abbreviations) and adversarial inputs (out-of-scope regions).

Success is defined by three primary metrics:

1. **Parsing Accuracy (F1 Score):** The system's ability to correctly extract and map entities (Street Name, Number, RT/RW) to the target JSON schema.
2. **Geospatial Validity Rate:** The percentage of outputs where the hierarchical chain (*Kelurahan* $\rightarrow$ *Kecamatan* $\rightarrow$ *Kota*) perfectly matches the Ground Truth database.

3. **Rejection Rate (Safety):** The system's efficacy in identifying and nullifying out-of-domain inputs (e.g., non-Jakarta addresses) to prevent hallucination.

6. ROADBLOCKS

6.1 The Vision

Awalnya, hipotesis kami: kalau SLM kami *fine-tune* pakai data alamat Jakarta* bakal langsung bisa dipakai & bisa sort out kasus-kasus seperti abbreviation, random spaces, alamat tidak lengkap dan acak, tetapi pas kami coba di preliminary testing, ternyata hasilnya gagal di hal-hal berikut:

- a. **Halusinasi**, ketika kami memasukan alamat yang diluar scope awal (jakarta) misal alamat Depok, model akan halusinasi dan bilang bagian dari jaksel. Atau kadang membuat nama kecamatan fiktif yang tidak ada di peta.
- b. **Random spaces**, masih kesusahan menangani random spaces seperti “Raja wati” (Rawajati)
- c. **Singkatan**, penanganan singkatan saat ini terbatas pada pola yang telah dipelajari model selama training. Oleh karena itu, ada possibility bahwa variasi singkatan unik atau jarang yang spesifik untuk wilayah Jakarta belum sepenuhnya ada dalam training dataset.
- d. **Typos**

6.2 Pivoting

Jadi, untuk menangani ketiga kendala utama tersebut (and many minor ones), kami memutuskan untuk beralih dari pendekatan ke **Hybrid Inference Pipeline**. Jadi hasil ekstraksi dari SLM tidak langsung diterima sebagai output final, melainkan divalidasi ulang oleh modul eksternal yang terhubung dengan knowledge base pemetaan wilayah Jakarta.

- a. **Handling hallucinations using database validator**, pada implementasi awal validator, kami menggunakan metode **exact match** ke database. Hasilnya sangat signifikan dalam mengurangi **halusinasi geografis**; entitas fiktif atau wilayah di luar Jakarta (seperti "Depok") berhasil ditolak. Namun, metode ini masih memiliki kelemahan, masih terlalu kaku dan gagal mengenali input yang valid namun memiliki *random spaces* atau *typo* (contoh: "Ra wa jati" ditolak karena tidak sama persis dengan "Rawajati").
- b. **Random spaces & typo**, untuk mengatasi kekakuan tersebut, kami menyempurnakan validator dengan mengintegrasikan **fuzzy matching**. Algoritma ini memungkinkan sistem untuk mencari data wilayah yang "paling mirip" secara matematis, bukan hanya yang sama persis. Namun kami belum melihat hasil yang begitu signifikan.

Tetapi di skenario pengujian dengan input alamat yang relatif bersih (hanya ngacak), model mampu melakukan entity extraction dengan baik.

6.3 Current Roadblocks

- a. Should we improve the dataset & generate more variative data? Or should we focus on the SLM itself? Or the validator/fuzzy? Try a different model?
- b. Random spaces, abbreviations, typos**

**Kami menggunakan data sintesis karena dua kendala utama: (1) ketatnya regulasi privasi terkait Personally Identifiable Information (PII) yang membatasi penggunaan data alamat riil pengguna, dan (2) kelangkaan dataset publik yang memuat variasi bahasa informal (slang) Jakarta secara memadai.*

*Untuk menghasilkan dataset ini, kami menerapkan **reverse-generation**, kami memulai dari data referensi (ground truth dataset mapping jakarta), kemudian secara sistematis 'mendegradasi' kualitasnya melalui **stochastic noise injection**. Proses ini meliputi pengubahan nama wilayah baku menjadi singkatan lokal, typos, serta pengacakan struktur alamat, sehingga menghasilkan data latih yang realistis, kaya variasi, namun sepenuhnya aman dari isu privasi.*

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nitip

colab:  PoC_prethesis.ipynb

github dataset sekunder: <https://github.com/teguh02/Wilayah-Indonesia-Beserta-Kode-Pos>

kumpulan dataset yang digunakan:  Datasets for Thesis

Reference Link (RAW)

1. <https://aclanthology.org/2024.emnlp-industry.29.pdf>
2. <https://aclanthology.org/2023.nlposs-1.3.pdf>
3. <https://arxiv.org/pdf/2410.20011> (slm yg kt pakai tuh apa ya) **Llama-3-8B-Instruct ini ya?**
4. <https://aclanthology.org/2025.acl-demo.42.pdf>
5. <https://dl.acm.org/doi/pdf/10.1145/3690624.3709425> kurang cocok sih
6. <https://arxiv.org/pdf/2409.11547> slm vs llm
7. <https://arxiv.org/pdf/2411.03350>
8. <https://dl.acm.org/doi/pdf/10.1145/3615890.3628538> address parsing
9. <https://github.com/teguh02/Wilayah-Indonesia-Beserta-Kode-Pos> dataset